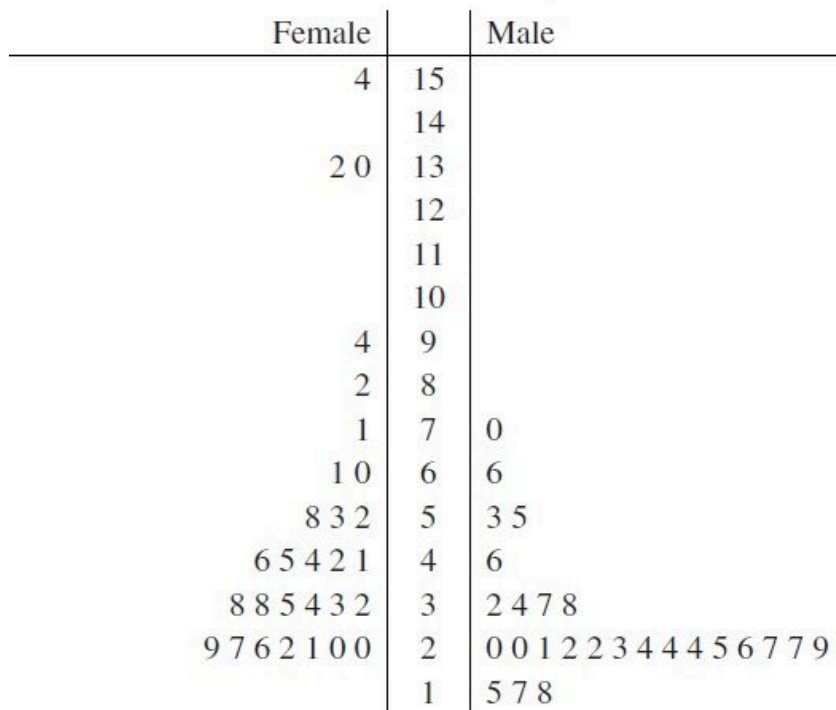


0909 U1 quantitative variable MCQ

1. In northwest Pennsylvania, a zoologist recorded the ages, in months, of 55 bears and whether each bear was male or female. The data are shown in the back-to-back stemplot below.



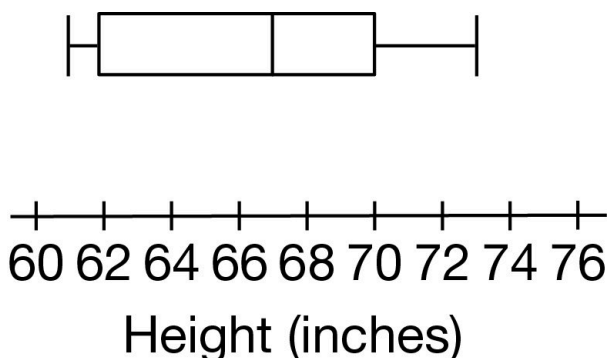
7|0 represents 70 months

Based on the stemplot, which of the following statements is true?

- (A) The median age and the range of ages are both greater for female bears than for male bears. ✓
- (B) The median age and the range of ages are both less for female bears than for male bears.
- (C) The median age is less for female bears than for male bears, and the range of ages is greater for female bears than for male bears.
- (D) The median age is greater for female bears than for male bears, and the range of ages is less for female bears than for male bears.

0909 U1 quantitative variable MCQ

2. The following boxplot summarizes the heights of a group of people who participate in a weekend biking club.



Which of the following statements is supported by the boxplot?

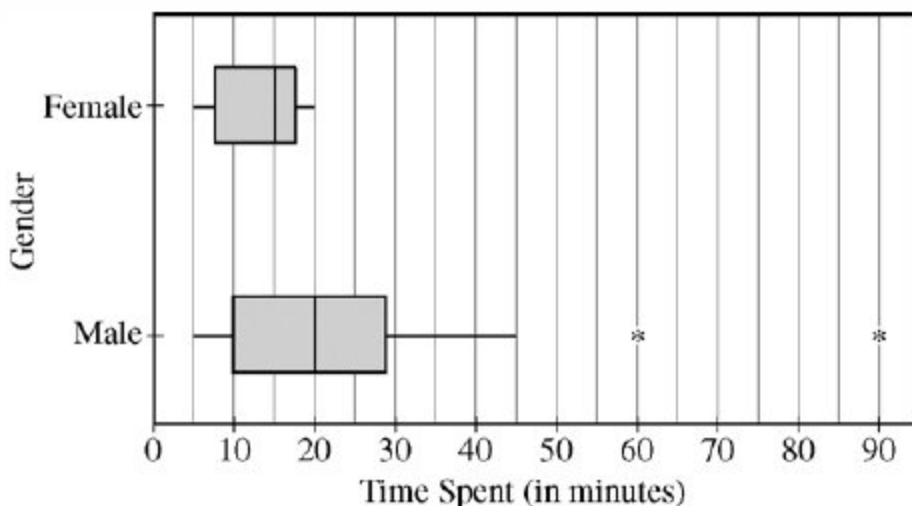
- (A) The mean height is 67 inches.
- (B) The number of people with height at least 70 inches is greater than the number of people with height at most 62 inches.
- (C) The number of people with height at least 67 inches is less than the number of people with height at most 67 inches.
- (D) Approximately 50% of the people have a height between 62 inches and 70 inches. ✓

Answer D

Correct. The interquartile range accounts for about 50% of the data. The first quartile is located at 62 and the third quartile is located at 70. Therefore approximately 50% of the people have a height between 62 inches and 70 inches.

0909 U1 quantitative variable MCQ

3. Janelle collected data on the amount of time in minutes each person in a large sample of customers spent in a local store. The data also included recording the gender of each customer. These data were used to generate the boxplots shown below.



Which of the following statements is true?

- (A) The range in the amount of time in minutes males in the sample of customers spent in the store is approximately 40 minutes.
- (B) The mean amount of time in minutes males in the sample of customers spent in the store is approximately 20 minutes.
- (C) The interquartile range of the amount of time in minutes females in the sample of customers spent in the store is 15 minutes.
- (D) Approximately half of the males in the sample of customers spent at least as much time in the store as any female in the sample of customers. ✓

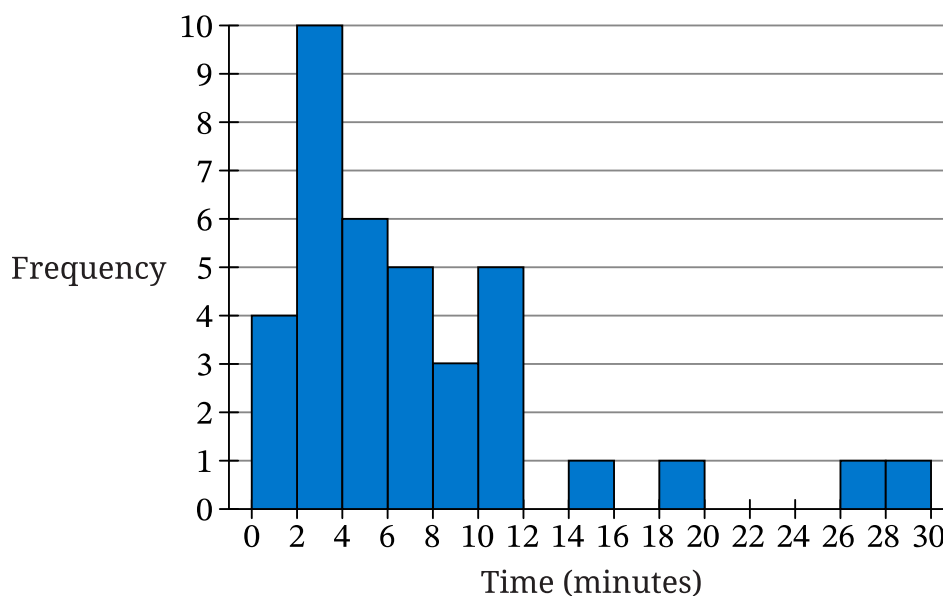
0909 U1 quantitative variable MCQ

4. During a 90-minute lecture, a teaching assistant recorded the amount of time, in minutes, that each student spent looking at their cell phone. A table displaying some summary statistics and a histogram displaying the results are shown.

Summary Statistics

n	Mean	Standard Deviation	Q1	Q3
37	6.865	6.447	2	9.5

Time Spent Looking at Cell Phone



Based on the information given and the $1.5 \times \text{IQR}$ rule, which of the following conclusions can be made about the median and the number of outliers?

- (A) The median is greater than 6.865, and the distribution has exactly two outliers.
- (B) The median is greater than 6.865, and the distribution has exactly four outliers.
- (C) The median is less than 6.865, and the distribution has exactly two outliers. ✓
- (D) The median is less than 6.865, and the distribution has exactly four outliers.

Answer C

0909 U1 quantitative variable MCQ

Correct. The median is less than the mean, 6.865, because the histogram's distribution is skewed to the right. The number of outliers is correct. The IQR is $Q3 - Q1$ which equals 7.5; $1.5 \times \text{IQR}$ results in 11.25. To find the lower outliers, 11.25 is subtracted from $Q1$. Because the lowest possible value is 0, there are no lower outliers. To find the upper outliers, 11.25 is added to $Q3$. There are two values above 20.75; one in the bin 26 to less than 28 and one in the bin 28 to less than 30. These two values are two upper outliers.

5. Suppose a new regulation requires businesses to pay all employees a minimum wage of \$15.00 per hour. A small coffee shop currently has 9 employees—5 employees whose wages are less than \$15.00 per hour and 4 employees whose wages are more than \$15.00 per hour. To meet the new regulation, the coffee shop plans to increase the wages of the 5 employees whose wages are less than \$15.00 per hour to \$15.00 per hour. The other 4 employees will not receive a wage increase at this time.

Which of the following must be true about the median wage of the 9 employees after the wage increase occurs?

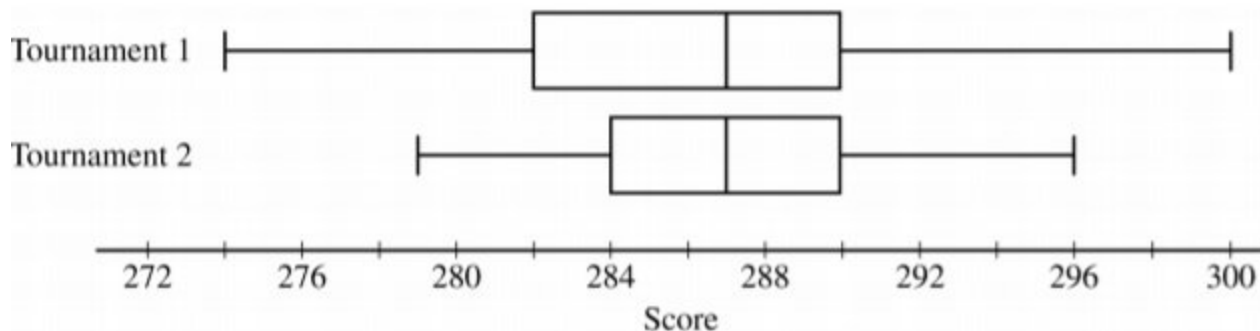
- (A) The median wage will be greater than \$15.00 per hour.
- (B) The median wage will increase but will be less than \$15.00 per hour.
- (C) The median wage will be \$15.00 per hour. ✓
- (D) The median wage will not change.

Answer C

Correct. The median of a set of data is the middle value when the data are ordered. If the coffee shop plans to increase the wages of the 5 employees whose wages are less than \$15.00 per hour to \$15.00 per hour, there will be 5 employees paid \$15.00 per hour and 4 employees paid more than \$15.00 per hour. The 5th value of the ordered data, the middle value, will be \$15.00 per hour.

0909 U1 quantitative variable MCQ

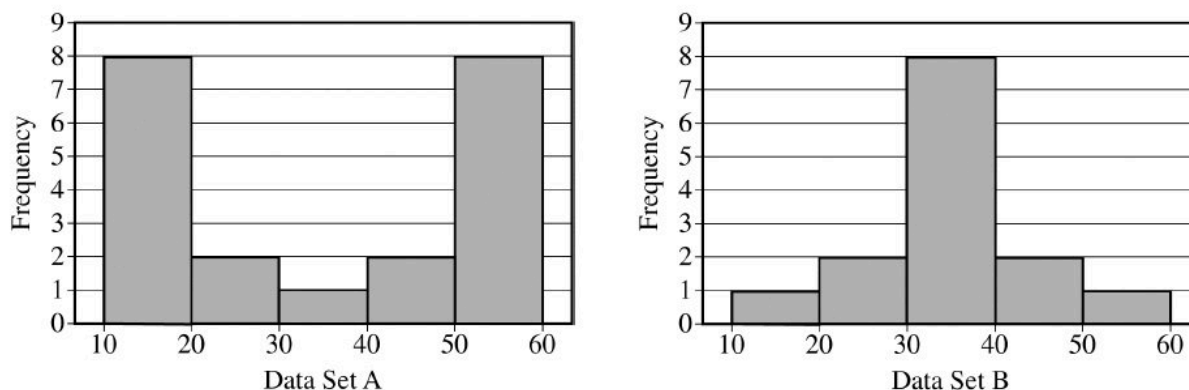
6. In a standard golf tournament, golfers play 18 holes of golf on each of 4 consecutive days. For each hole, golfers keep track of the number of times they hit the ball (strokes) before the ball goes into the cup. A golfer's score for the tournament is the total number of strokes needed to complete the tournament. The boxplots below summarize the scores for golfers who competed in tournament 1 and golfers who competed in tournament 2.



Based on the boxplots, which of the following statements must be true?

- (A) More golfers played in tournament 1 than in tournament 2.
- (B) In both tournaments, at least half the golfers completed the tournament with a score less than 288. ✓
- (C) The number of golfers who completed tournament 1 with a score less than 288 was greater than the number of golfers who completed tournament 2 with a score less than 288.
- (D) The score of the golfer with the least score in tournament 1 was greater than the score of the golfer with the least score in tournament 2.

7.

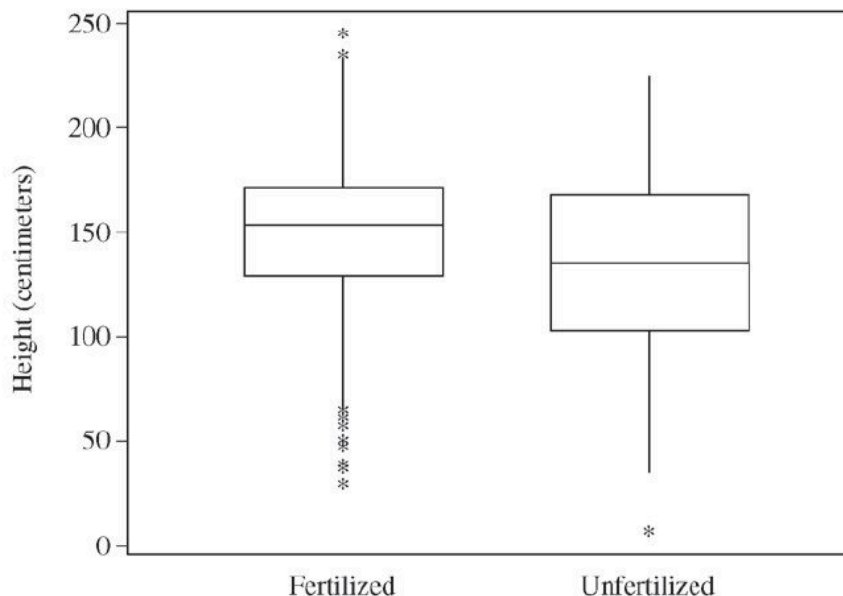


Which of the following statements must be true about the data sets A and B displayed in the histograms above?

- (A) The median of data set A is equal to the median of data set B.
- (B) The range of data set A is equal to the range of data set B.
- (C) The standard deviation of data set A is less than the standard deviation of data set B.
- (D) The standard deviation of data set A is greater than the standard deviation of data set B. ✓

0909 U1 quantitative variable MCQ

8.



The figure above summarizes the heights, in centimeters, of approximately 400 pine seedlings six years after they were planted at a center for environmental study. Approximately half of the trees were fertilized yearly, and the remaining trees were never fertilized. Which of the following statements about the medians and interquartile ranges (IQRs) of the heights of the two groups of trees 6 years after being planted is true?

- (A) The median for the unfertilized trees is greater than the median for the fertilized trees, and the IQR is also greater for the unfertilized trees.
- (B) The median for the unfertilized trees is the same as the median for the fertilized trees, and the IQR is greater for the unfertilized trees.
- (C) The median for the unfertilized trees is less than the median for the fertilized trees, and the IQR is greater for the unfertilized trees. ✓
- (D) The median for the unfertilized trees is less than the median for the fertilized trees, and the IQR is less for the unfertilized trees.

9.

DESCRIPTIVE STATISTICS

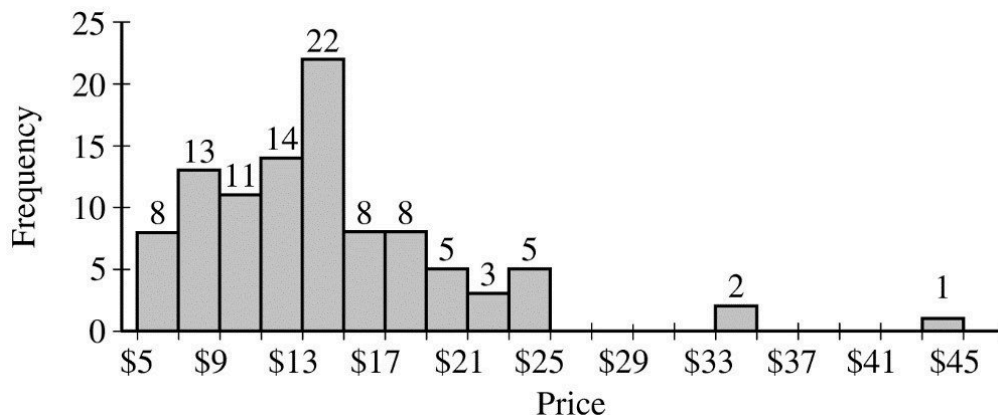
Variable score	N	Mean	Median	TrMean	StDev	SE Mean
	50	1045.7	1024.7	1041.9	221.9	31.4
Variable score	Minimum	Maximum	Q1	Q3		
	628.9	1577.1	877.7	1219.5		

Some descriptive statistics for a set of test scores are shown above. For this test, a certain student has a **standardized score of** $z = -1.2$. What score did this student receive on the test?

- (A) 779.42 ✓
- (B) 1008.02
- (C) 1083.38
- (D) 1311.98

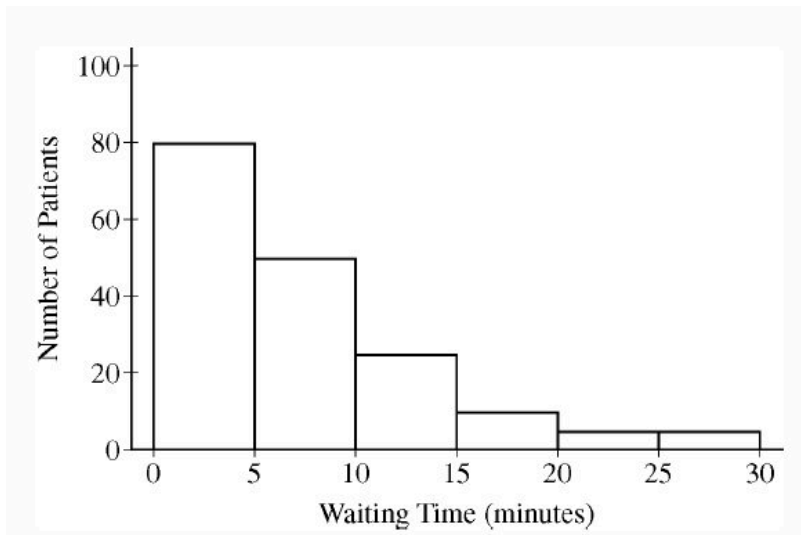
0909 U1 quantitative variable MCQ

10. The histogram shown summarizes the responses of 100 people when asked, “What was the price of the last meal you purchased?”



Based on the histogram, which of the following could be the interquartile range of the prices?

- (A) \$21
- (B) \$10
- (C) \$5 ✓
- (D) \$3
11. The histogram below displays the frequencies of waiting times, in minutes, for 175 patients in a dentist's office.



Which of the following could be the median of the waiting times, in minutes?

- (A) 2.50
- (B) 7.25 ✓
- (C) 12.25
- (D) 15.00

0909 U1 quantitative variable MCQ

12. The caffeine content of 8-ounce cans of a certain cola drink is approximately normally distributed with mean 33 milligrams (mg). A randomly selected 8-ounce can containing 35 mg of caffeine is 1.2 standard deviations above the mean. Approximately what percent of 8-ounce cans of the cola have a caffeine content greater than 35 mg?
- (A) 1%
(B) 8%
(C) 12% ✓
(D) 16%
13. A grocery store received three shipments of bags of cranberries: one from farm A, one from farm B, and one from farm C. The distributions of the weight of a bag are unimodal and symmetric for all three shipments. The number of bags and the mean and standard deviation, in ounces (oz), for each shipment are shown in the table.

Farm	Number of Bags	Mean (oz)	Standard Deviation (oz)
A	100	16.0	0.5
B	100	16.0	0.1
C	100	16.0	0.1

Which of the following statements is true?

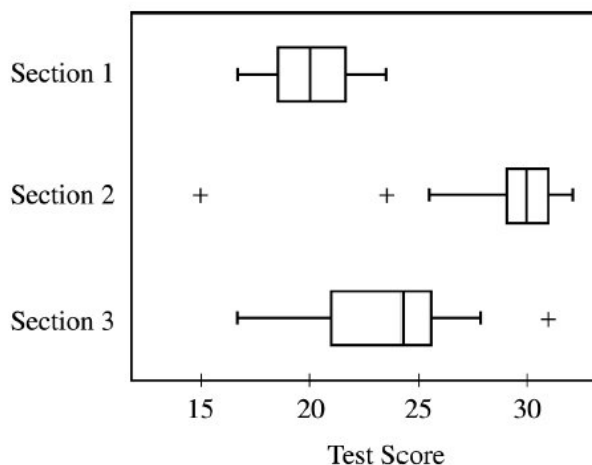
- (A) The shipment from farm A is most likely to have the greatest number of bags that weigh less than 16 oz.
(B) The shipment from farm B is most likely to have the greatest number of bags that weigh less than 16 oz.
(C) The shipments from farms B and C are the only shipments that are likely to have an approximately equal number of bags that weigh less than 16 oz.
(D) The shipments from farms A, B, and C are all likely to have an approximately equal number of bags that weigh less than 16 oz. ✓

Answer D

Correct. The distributions of the weight of a bag are unimodal and symmetric for all three shipments, and each shipment has a mean weight of 16 ounces. Therefore, each shipment should have approximately half of the bags (50) less than 16 ounces.

0909 U1 quantitative variable MCQ

14.



Professor James gave the same test to his three sections of statistics students. On the 35-question test, the highest score was 32 and the lowest was 15. Based on the information displayed in the boxplots above, which of the following statements is true?

- (A) Section 1 has the smallest interquartile range.
- (B) Section 2 has the smallest range of scores.
- (C) The top 25% of scores in section 2 are lower than the highest score in section 3.
- (D) At least 50% of the scores in section 3 are higher than all of the scores in section 1. ✓

15. Each person in a random sample of adults was asked how many DVDs he or she owned. Summary statistics are given below.

Variable	N	Mean	Median	TrMean	StDev	SE Mean
DVDs	117	129.4	50.0	76.5	323.6	29.2
Variable	Minimum		Maximum		Q1	Q3
DVDs	0.0		3000.0		30.0	95.0

Which of the following statements is true?

- (A) Seventy-five percent of the adults in the sample own more than 95 DVDs.
- (B) Fifty percent of the adults in the sample own between 0 and 129.4 DVDs.
- (C) The interquartile range of the number of DVDs owned is 65. ✓
- (D) The distribution of the number of DVDs owned contains outliers on both the low side and the high side.

16. Each value in a sample has been transformed by multiplying by 3 and then adding 10. If the original sample had a variance of 4, what is the variance of the transformed sample?

- (A) 4
- (B) 12
- (C) 22
- (D) 36 ✓

0909 U1 quantitative variable MCQ

17. Height, in meters, is measured for each person in a sample. After the data are collected, all the height measurements are converted from meters to centimeters by multiplying each measurement by 100. Which of the following statistics will remain the same for both units of measure?
- (A) The mean of the height measurements
- (B) The median of the height measurements
- (C) The standard deviation of the height measurements
- (D) The z-scores of the height measurements ✓
18. A local real estate magazine used the median instead of the mean when it reported the SAT score of the average student who attends Groveland High School. A graphical display of SAT scores of students who attend Groveland High School indicated that the data were strongly skewed to the right. Which of the following explains why, in this situation, the median is a more accurate indicator of the SAT score of the average student than the mean is?
- (A) The mean is affected by the skewness, whereas the median is not. ✓
- (B) The mean will be less than the median when the data are strongly skewed to the right.
- (C) The mean should be used only when data are strongly skewed to the left.
- (D) The median is equal to one-half the sum of the maximum and minimum SAT scores at Groveland High.
19. A golfer recorded the following scores for each of four rounds of golf: 86, 81, 87, 82. The mean of the scores is 84. What is the sum of the squared deviations of the scores from the mean?
- (A) $\sum(x - \bar{x}) = (86 - 84) + (81 - 84) + (87 - 84) + (82 - 84)$
- (B) $\sum|x - \bar{x}| = |86 - 84| + |81 - 84| + |87 - 84| + |82 - 84|$
- (C) $\sum(x - \bar{x})^2 = (86 - 84)^2 + (81 - 84)^2 + (87 - 84)^2 + (82 - 84)^2$ ✓
- (D) $[\sum|x - \bar{x}|]^2 = [|86 - 84| + |81 - 84| + |87 - 84| + |82 - 84|]^2$

Answer C

Correct. The formula for finding the variance of a sample is $s^2 = \frac{\sum(x - \bar{x})^2}{n-1}$, where $\sum(x - \bar{x})^2$ is the sum of the squared deviations from the mean. For the golf scores,

$$\sum(x - \bar{x})^2 = (86 - 84)^2 + (81 - 84)^2 + (87 - 84)^2 + (82 - 84)^2 = 4 + 9 + 9 + 4 = 26.$$

0909 U1 quantitative variable MCQ

20. For a random sample of 50 newborn Holstein calves from the United States, the mean weight was 80 pounds and the standard deviation was 6 pounds. In order to compare these findings with those of other countries, an agricultural engineer converted the measured calf weights from pounds to kilograms by multiplying each weight by 0.454.

What are the mean and standard deviation of the converted weights of the 50 newborn Holstein calves?

- (A) The mean weight is 36.32 kilograms and the standard deviation is 2.724 kilograms. ✓
- (B) The mean weight is 36.32 kilograms and the standard deviation is 6 kilograms.
- (C) The mean weight is 80 kilograms and the standard deviation is 2.724 kilograms.
- (D) The mean weight is 80 kilograms and the standard deviation is 6 kilograms.

Answer A

Correct. Changing the units of measurement affects the values of the calculated statistics. When each measurement is multiplied by a constant, the sample mean and sample standard deviation are both multiplied by that same constant to get the sample mean and sample standard deviation in the new units. The mean weight is $80(0.454) = 36.32$ kilograms, and the standard deviation is $6(0.454) = 2.724$ kilograms.